Bioactive molecules in thermal waters: A revolution in cosmetic products

Zeina BOURHANE*, Cristiana Cravo-Laureau, Marine Meslier, Hervé Pinaly, Robert Duran, Hugues Preud'homme

Université de Pau et des Pays de l'Adour, UPPA, CNRS, UMR5254 - IPREM, 64000 Pau, France

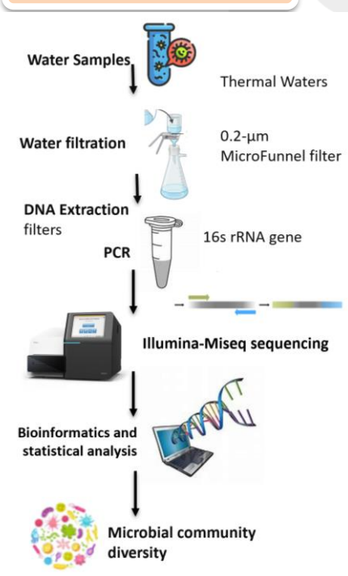
Introduction

Pyrenees region is rich in thermal water with recognized therapeutic properties. These sulfurous waters are distinguished by their varied physical and chemical characteristics, as well as the presence of ions and minerals with interesting effects. However, the microbiome and bioactive metabolites present in these waters and their possible effects on the skin are unknown. Our aim is to characterize the microbial community and their relative metabolites in different thermal waters in the Pyrenees. Finally, the transfer of this scientific knowledge to local cosmetic companies will help develop innovative dermo-cosmetic products.

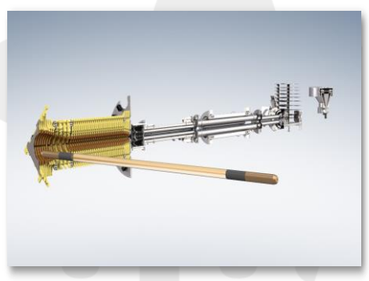
Methodology

- 11 Thermal water sampling sites across the Pyrenees region were investigated

1-Microbial analysis



2- Metabolomic analysis



Non targeted analysis with a Brucker® tims-TOF with the following specificities:

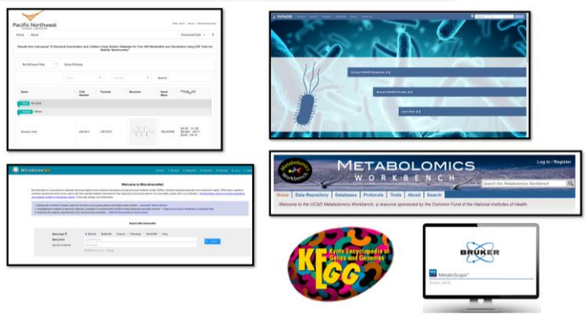
- Reversed geometry (the gas (N_2) is blowing in the same direction as molecular ion elution)
- Trapped ion mobility: tunable resolution with duty cycle around 100% & ability to trap ion cloud.

3- Ions & Minerals analysis

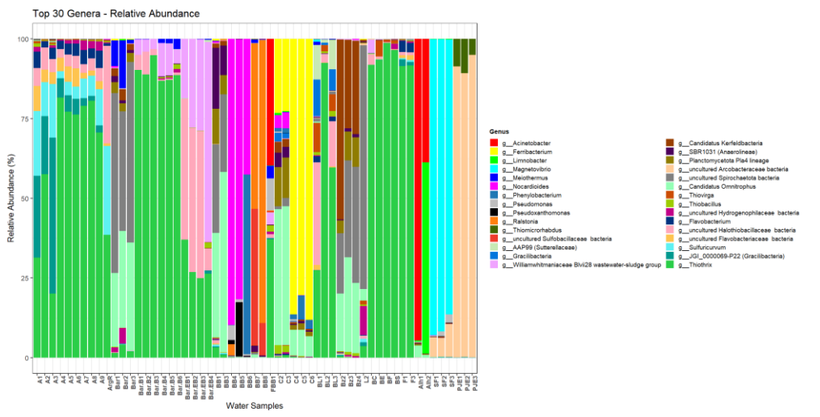
4- Phys-Chem parameters

pH, salinity, Conductivity , Dissolved oxygen, Temp

5- Metabolites-microbiome correlation

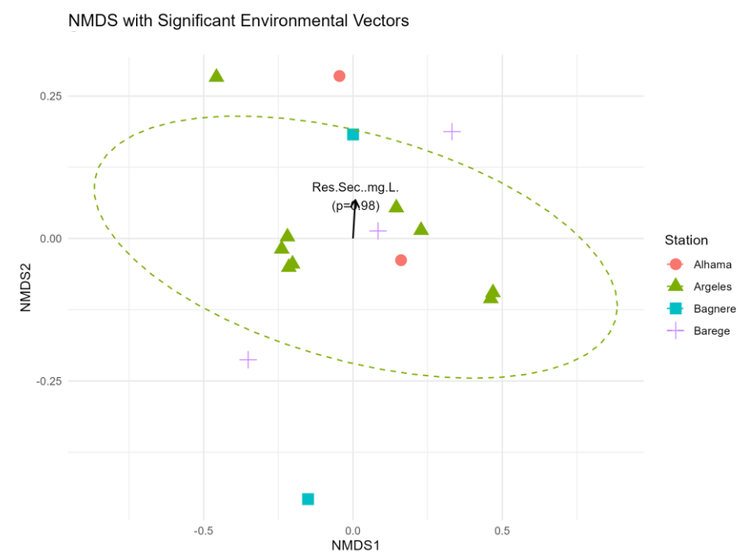


Prokaryotic community composition



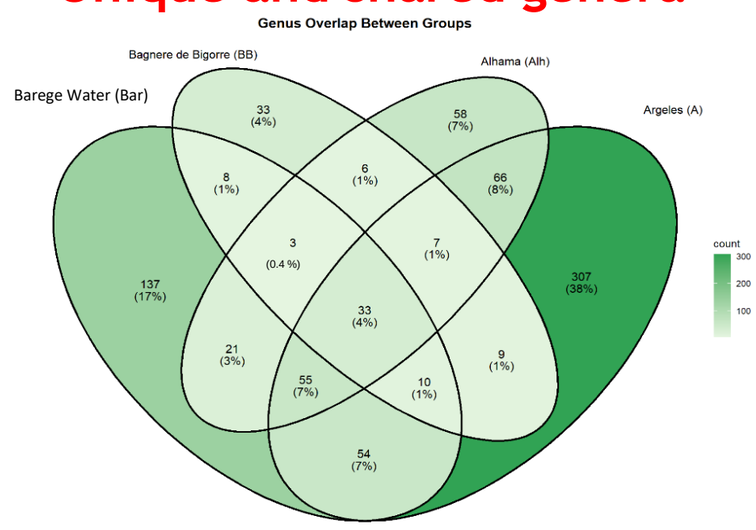
- The microbial communities of thermal water samples were dominated by sulfur-oxidizing bacteria affiliated to *Thiothrix*, *Sulfuricurvum* and *Flavobacterium* genus.

Microbial community distribution



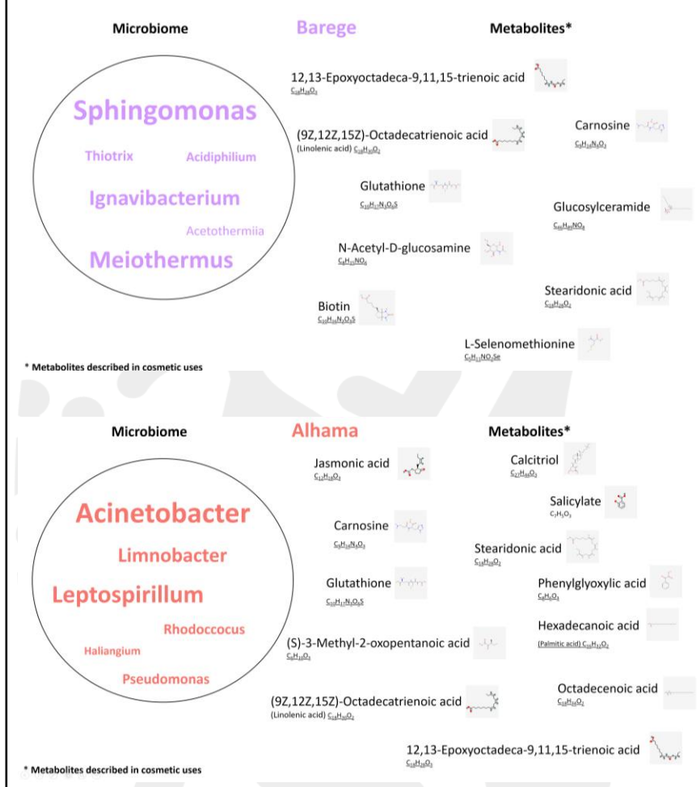
- The comparison of the microbial communities by NMDS separated the Stations in different clusters, highlighting their different microbial composition.
- The NMDS revealed that the *res sec* influence significantly the microbial communities of Alhama and Bagnere waters.

Unique and shared genera



- The Venn diagram revealed a core microbial community shared among the four sites, representing 4% of the total taxa.
- In addition, several genera were unique to each thermal site, highlighting site-specific microbial signatures.

Microbiome-Metabolites correlation



- Although numerous metabolites were identified, only those with documented pharmaceutical or cosmetic applications were considered and linked to specific microbial genera.
- The characterized metabolites belong to the sterol, organic acid, and fatty acid families.
- Glucuronic acid was associated with the *Flavobacterium* genus, phenylglyoxylic acid with *Bacillus*, and hexadecanoic acid with *Thiothrix*.

Conclusion

- Pyrenean thermal waters harbor a unique and site-specific microbiome, with metabolites of anti-inflammatory and therapeutic potential linked to specific microbial genera. These findings highlight the promise of bioactive metabolites from thermal waters for innovative applications in cosmetic formulations.
- Our study provides the scientific knowledges for this revolution, demonstrating how unique microbiomes and their bioactive metabolites can be harnessed to develop innovative and sustainable cosmetic solutions.

Acknowledgment

CARUSO project EFA085/01 (€1,054,799) is 65% co-financed by the European Union through the Interreg VI-A Spain-France-Andorra program (POCTEFA 2021-2027). The objective of POCTEFA is to strengthen the economic and social integration of the Spain-France-Andorra border area.